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Investigating the prevalence and utility of police body-worn cameras in the George Floyd protests

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Received 19 October 2021
Revised 6 January 2022
9 March 2022
Accepted 25 April 2022

Abstract

Purpose – Unlike protests against police brutality in the past (2014 and earlier), police officers responding to First Amendment-protected demonstrations in summer 2020 likely were wearing body-worn cameras(BWCs). This study seeks to understand police perceptions of the effects of BWCs when used in the George Floyd protests.

Design/methodology/approach – The authors use survey data from 100 agencies with federally-funded BWCs to assess the prevalence of BWC deployment to George Floyd protests and perceived benefits and limitations of the technology within this unique context.

Findings – About three-quarters of agencies encountered some level of demonstration/protest related to the killing of George Floyd, and the majority of those deployed BWCs during these demonstrations. Respondents indicated evidentiary value of footage was a key reason for doing so, and at least three preconditions for a civilizing effect were present.

Originality/value – Research has documented numerous benefits associated with BWCs, from reductions in use of force and citizen complaints to evidentiary value. However, the extent to which BWC benefits extend to public protests is unclear. The George Floyd protests represent an opportunity to understand the prevalence and usefulness of BWCs in policing public protests.

Keywords Police, Body-Worn Cameras (BWCs), Protests, Black lives matter, Civilizing effect

Paper type Research paper

Introduction

In the United States, Black citizens have long suffered the disproportionate impact of police abuse of authority, from discrimination in "stop-and-frisk" encounters (Gelman *et al.*, 2007) to non-lethal (e.g. Fryer, 2020; Geller *et al.*, 2020) and lethal use of force (e.g. Campbell *et al.*, 2018; Fyfe, 1988). White and Fradella (2016, p. 9) argue there is a persistent undercurrent of "racial injustice and discrimination that has served as a backdrop to professional policing in the

The authors thank the editors and anonymous reviewers for their helpful comments. Additionally, they thank the survey respondents for their time and candor. This research was carried out with funding support from the Bureau of Justice Assistance (BJA), U.S. Department of Justice. The opinions expressed here are those of the authors and do not necessarily reflect the official position or policies of BJA or the U.S. Department of Justice.

Conflict of interest: This project was completed with support from our partners at CNA. Dr. White has a consulting relationship with compensation from CNA.

Policing: An International Journal
Vol. 45 No. 4, 2022
pp. 633-647
© Emerald Publishing Limited
1363-951X
DOI 10.1108/PIJPSM-10-2021-0151

United States for the last 175 years." Police killings of citizens are the most contentious and extreme form of police authority. White (2006, p. 303) notes shootings "are controversial even under the best circumstances, and they can have potentially devastating consequences not only for the victim and the officer but also for the police department, the community, and their relationship." This pattern of controversial use of police authority has continued unabated into the 21st century. In 2014–15, the deaths of Michael Brown, Freddie Gray, Walter Scott, and others led to public outrage, protests and riots, and demands for police reform.

In response, former President Obama created the President's Task Force on 21st Century Policing, and the final report (2015) identified nearly 60 recommendations for building public trust and enhancing police accountability. Yet five years later, Minneapolis (MN) police killed George Floyd, Louisville (KY) police killed Breonna Taylor, Atlanta (GA) police killed Rayshard Brooks, and Kenosha (WI) police shot Jacob Blake (White *et al.*, 2021). Again, there were protests and public outrage.

Though images from protests against police brutality in summer 2020 were similar to images from Ferguson and Baltimore in 2014–2015, Los Angeles in 1992, and many other cities in the 1960s, the George Floyd protests in 2020 [1] were marked by two important differences. The first is the widespread, organized nature of the protests which extended far beyond previous police reform movements (Silverstein, 2021). The Black Lives Matter (BLM) movement grew significantly after Michael Brown was killed in 2014. However, after George Floyd's death, BLM "established itself as a worldwide movement . . . [and] spearheaded demonstrations worldwide protesting police brutality and systematic racism" (Howard University Law Library, 2021). These BLM-led protests were accompanied by professional athletes boycotting games (Hill, 2021); proposed police reform legislation at the local, state, and federal levels (White *et al.*, 2021); and calls to defund the police (Eaglin, 2021; Weichselbaum and Lewis, 2020).

The second difference involves the presumed presence of police body-worn cameras (BWCs) at the George Floyd protests. In 2014–15, law enforcement agencies moved swiftly to deploy BWCs. By 2016, about half of all agencies—and 80% of large agencies with more than 500 sworn personnel—were using them (Hyland, 2018; Nix *et al.*, 2020). However, the extent to which agencies deployed BWCs to protests remains unclear, as civil rights groups have opposed their use at First Amendment-protected demonstrations (Stanley, 2020). Also unclear is the perceived usefulness of BWCs at protests, both in terms of influencing behavior and by capturing footage with evidentiary value. In other words, do the positive effects of BWCs extend to policing public protests?

This research gap is notable given the potential for BWC footage to document events in real-time and provide a source of information about the actions of both police and citizens. Our study addresses the potential for the documented positive effects of BWCs to extend to police-citizen interactions during public protests through a survey of 100 agencies receiving federal funds for BWCs. The survey captures the extent to which BWCs were deployed at George Floyd protests and agency perceptions of both the technology's value and limitations in this context.

Literature review

Police BWCs and behavior change

Police BWCs had been used by agencies in Canada and the United Kingdom for over a decade before widespread diffusion in America began in 2014–2015 (Hyland, 2018; Nix *et al.*, 2020). Though this occurred before a solid research base on BWCs had been established (Lum *et al.*, 2019; White and Malm, 2020), researchers quickly responded. As of January 2020, there were 119 studies on the various effects of police BWCs (Gaub and White, 2020), with that number steadily growing (e.g. Braga *et al.*, 2022).

Researchers have focused significant attention on the effect of BWCs on officer use of force [2] and citizen complaints. Early on, most studies found BWCs led to decreased officer use of force (e.g. Ariel *et al.*, 2015; Jennings *et al.*, 2015). Some later studies confirmed these

findings (e.g. Jennings *et al.*, 2017; Koslicki *et al.*, 2020; White *et al.*, 2018), whereas others found no effect (e.g. Braga *et al.*, 2018; Gaub *et al.*, 2021b; Peterson and Lawrence, 2020; Yokum *et al.*, 2017). To date, 14 of 28 studies document notable or statistically significant declines in use of force (White *et al.*, 2021b). The findings related to citizen complaints have been more consistent, with the majority of studies (24 of 32; White *et al.*, 2021a) finding significant declines in overall citizen complaints following the implementation of BWCs (e.g. Hedberg *et al.*, 2017; Jennings *et al.*, 2017, 2015; Peterson and Lawrence, 2020) [3].

Two theories explain improved citizen and officer behavior as a result of being recorded on BWC, often referred to as a "civilizing effect" (White and Malm, 2020). The first is self-awareness theory (Duval and Wicklund, 1972), which "suggests that when human beings are under observation, they modify their behavior, exhibit more socially acceptable behavior . . . and cooperate more fully with the rules" (Braga *et al.*, 2018, p. 516). The second is deterrence theory, which states that swift, certain, and severe punishment will induce people to avoid negative behavior (Akers, 1990; Beccaria, 1963). Though each theory more broadly has considerable empirical support, the extent to which either directly explains BWC-driven reductions in use of force and citizen complaints remains uncertain. Patterson and White (2021) identified four pre-conditions that must be met in order for a civilizing effect to occur: awareness of the camera, activation of the camera, escalated behavior (or the potential for it), and the mental capacity to weigh the consequences of being recorded (e.g. rational thought not impaired by drugs, alcohol, mental illness, etc.). While these pre-conditions are nearly always present for police officers, research suggests all four may rarely be met for citizens (Patterson and White, 2021). Nonetheless, this does not always preclude officers from recording themselves engaging in wrongdoing. For example, former Jackson County (FL) sheriff's deputy Zachary Wester planted drugs during countless traffic stops (Ciaramella, 2021) and former Louisiana State Trooper Chris Hollingsworth's use of force resulted in the death of Ronald Greene (Almasy *et al.*, 2021)—both of which were captured on BWC.

Police BWCs and evidentiary value

The evidentiary value of BWCs (i.e. using footage as evidence in police investigations or in court proceedings) has received considerably less attention (White and Malm, 2020). Harris (2010) suggested the presence of BWCs may increase officers' compliance with rules (e.g. search and seizure, Miranda warnings). BWC footage could also lead to more accurate report writing by officers (Boivin and Gendron, 2021; Vredeveldt *et al.*, 2021). Several studies have found that BWCs can influence the investigation of citizen complaints, including substantial cost savings (Braga *et al.*, 2017; Katz *et al.*, 2014).

A handful of studies have examined the impact of BWCs on criminal case processing, wherein it is assumed that better case-processing outcomes (e.g. faster case resolution, increased rate of guilty pleas) result from BWC evidence. Both Owens *et al.* (2014) and Morrow *et al.* (2016) reported that BWCs led to enhanced outcomes in domestic violence cases (greater likelihood of arrest, charging, and conviction). White *et al.* (2021c) found that BWCs had no impact on misdemeanor drug and alcohol case disposition, but did significantly reduce case-processing time. Alternatively, Yokum *et al.* (2017, p. 22) compared judicial outcomes among Washington, DC cases with and without BWC footage, and found no "clear benefits." Prior research has not tested whether BWCs have evidentiary value during protests, though the footage from such events would presumably be equally useful in criminal prosecutions of wrongdoing by both protesters and police.

All of the aforementioned benefits of BWCs have been identified for encounters between police and citizens under more everyday circumstances, such as calls for service. The extent to which any of these BWC-driven benefits extend to public protests is unknown, as protests are unique events that differ in significant ways from the average police-citizen encounter (Maguire and Oakley, 2020).

Social movements, group behavior, and protester dynamics

Protests and resulting social movements begin when social factors (e.g. politics, injustice) generate strong grievances against actions performed by government entities (Heaney and Rojas, 2007; Jenkins *et al.*, 2003). Social identity with oppressed groups is important for participating in protests (van Zomeren *et al.*, 2004), as is strong negative emotions (e.g. de Vos *et al.*, 2013; Iyer and Ryan, 2009). The link between group identity and experiencing unfairness is reciprocal, because experiencing unfairness or injustice often reinforces group identity and creates a sense of anger (van Zomeren *et al.*, 2004). Experiences that reinforce in-group identification coupled with negative emotions facilitates opportunities for collective action and inspires social movements and protest.

At the core of protest may be injustice, grievance, social identity, and negative emotionality, but behaviors are sometimes balanced with organization and planning to motivate group dynamics (Maguire and Oakley, 2020). Protests are sometimes organized around core ideas, and violent behavior is often only supported in the face of actual or perceived police unfairness during protests (Maguire *et al.*, 2018). Because protest behavior is different from individual behavior, the dynamics of police responses to protests are different than traditional policing (Maguire and Oakley, 2020). Therefore, BWCs may have different effects on officer and citizen behavior in protest settings.

Policing protests and police behavior

Research on protest policing ranges from the repression of civil disobedience (Marx, 1970) to more recent studies of police tactics and dynamics (e.g. the use of procedural justice [4]; see Snipes *et al.*, 2019; Tyler *et al.*, 2018). Research demonstrates the importance of understanding police behavior as a potential motivator of protester behavior (Snipes *et al.*, 2019; Tyler *et al.*, 2018). When police tactics are repressive, protestors are more inclined to behave violently (Maguire *et al.*, 2020, 2018). For those especially committed to their cause, police violence has no direct effect on deterring activism (Cobbina *et al.*, 2019). Repressive police behavior is highly indicative of protester violence and does little to prevent/stop a protest (Maguire, 2021). BWCs offer a unique opportunity to observe the behavior of officers and protesters and, this study explores the potential of BWCs to alter the interplay between the two.

Current study

Using a survey of police agencies that received federal funds to implement BWCs, we examine four research questions (RQs):

RQ1. How prevalent and civil were the George Floyd protests?

RQ2. How often were police BWCs present during those protests?

RQ3. What are the perceived benefits and drawbacks of using BWCs at those protests?

RQ4. Are there differences in perceptions based on agency size?

Data and methods

Data

We use data from a survey of agencies who received federal funding to equip their officers with BWCs through the Bureau of Justice Assistance BWC Policy and Implementation Program (Bureau of Justice Assistance, 2021). All grant recipients for fiscal years 2015–2019 (N = 426) [5] were invited to participate via email in late-June 2020. The survey asked respondents to describe their experience with protests specifically tied to the death of George Floyd and/or the Black Lives Matter movement that took place on or after May 25, 2020

(the day George Floyd was killed in Minneapolis, MN). The online Qualtrics survey included approximately 20 open- and closed-ended questions covering the prevalence of protests in the reporting agency's jurisdiction, the civility of the protests (measured as the agency's perception on a scale of 1–10; 1 = very peaceful, 10 = very violent), use of BWCs at those protests, key aspects of their use (e.g. activation), and the usefulness of BWCs [6]. Our questions focused on respondents' perceived use of BWCs to produce behavior change (i.e. "civilizing effect") and capture footage that has evidentiary value. We also asked respondents to describe any limitations or technical problems with BWCs while at the protests and the prevalence of citizen complaints filed against officers during protests. Of the 373 agencies invited to participate, we received 100 completed surveys (26.8% response rate). The sample included local police departments (n = 80), sheriff's departments (n = 15), and special jurisdiction agencies (n = 5). We separately captured information about each agency's official written policy on citizen notification (i.e. are officers required to tell citizens at a protest that they are being recorded?).

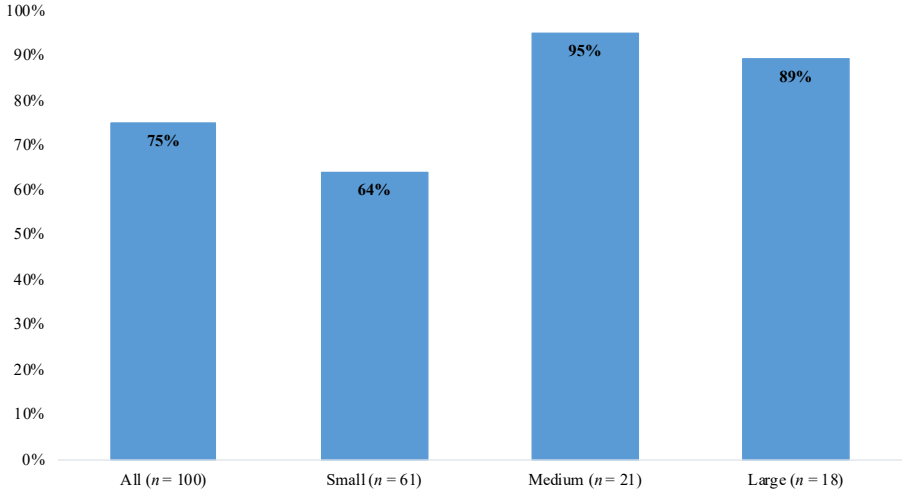
Analytical plan

Given the exploratory nature of the study, we rely on descriptive and bivariate analysis of the survey results. We also employed thematic analysis (Boyatzis, 1998; Ryan and Bernard, 2003) of open-ended questions. Survey results categorized and examined by agency size: Small (0–99 sworn officers; n = 61), medium-sized (100–499 sworn officers; n = 21) and large (500 or more sworn officers; n = 18) [7]. We examine differences by agency size using chi-square, one-way ANOVA (for civility which is measured as a mean), and Bonferonni post-hoc tests.

Results

Prevalence and civility of protests

Figure 1 displays the prevalence of protests, both overall (RQ1) and by agency size (RQ4). Altogether, 75 respondents reported that they experienced protests following the death of George Floyd. Of the 25 that did not experience protests, 22 were small (all had fewer than 50



Note(s): Pearson Chi(2) = 10.42, p-value < 0.001

Figure 1. Prevalence of reported protests by agency size

sworn officers) and two were correctional agencies. Though public protests occurred in most jurisdictions, they were significantly less common among small agencies (64%) compared to medium-sized (95%) and large (89%) agencies (chi-square = 10.42, $p < 0.001$). We also asked whether officers assisted with protests in other jurisdictions (not shown). This practice was common among small and medium-sized agencies (52% for each; large, 39%). Also, protest prevalence varied by agency type, with the majority occurring within local and county jurisdictions (local agencies, 76%; sheriff's departments, 80%; and special jurisdiction agencies, 40%) [8].

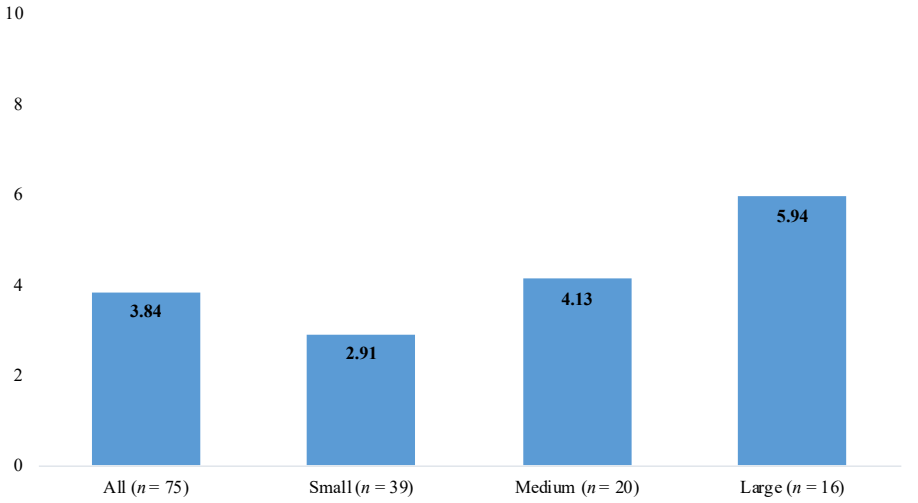
Agencies that experienced public protests were asked to rate the civility of the protests on a scale of 1–10 (1 = very peaceful, 10 = very violent; see Figure 2) to address the civilizing effect pre-condition of escalated behavior (Patterson and White, 2021). Protests were generally peaceful (mean = 3.84), though the average violence level increased with agency size (2.91, small; 4.13, medium; 5.94, large; $F = 10.49$, $p < 0.001$; only small v. large comparison reaches statistical significance). Several comments highlight this point:

> We have only experienced a few [protests] so far and all have been pretty peaceful. (Small agency)

> [Our] protests have been peaceful . . . We have had some type of protest/rally for 14 days now. (Small agency)

We also asked agencies to report on the prevalence and nature of officer injuries at protests (not shown). More than 80% of responding agencies stated that no officers were injured [9]. Injuries were more commonly reported by large agencies, with serious injuries in some cases. Of the 18 large agencies that responded, 10 (56%) reported that officers were injured at the protests.

> Multiple officers suffered superficial physical injuries (cuts, bruises, etc.) due to objects and debris thrown at them. One motorcycle officer was hit by an ATV speeding at over 70 mph, resulting in severe injury to his legs and threat of not being able to walk again. Another officer suffered a concussion during a use of force incident. (Large agency)



Note(s): Mean scores presented; ANOVA F-test = 10.49, p-value < 0.001

Figure 2.
Average civility of reported protests by agency size

> Yes [officers were injured]. Numerous officers were struck with bricks, rocks, bottles and ball bearings. Most common injuries were concussions and contusions. (Large agency)

Prevalence and usefulness of BWCs at protests

Table 1 highlights the prevalence and usefulness of BWCs at the protests by agency size (RQ2 and RQ4). BWCs were present at the majority of protests (nearly 90%) and BWC use was prevalent regardless of agency size (86% small, 82% medium-sized, 94% large). Overall, 78% of reporting agencies indicated that BWCs were useful at protests (there was no statistically significant difference by size). Small and medium-sized agencies that indicated BWCs were not useful most commonly stated they were not needed given the civility of their protests.

> As a result of our department's strong community outreach and relationship building, there was no urgent need for BWC footage during the two highly peaceful demonstrations that have taken place in City C. Much credit must be given to the organizers and participants in these protests, as well, in preventing any violent or destructive behavior. (Medium-sized agency)

> [D]ue to our soft approach to the protesters and the overall peaceful attitude of the protesters themselves; there was no need for audio/video recording by BWC, since the protest was peaceful, no arrests were made, and no criminal activity needed to be documented. (Medium-sized agency)

Perceived benefits and drawbacks

Respondents noted several perceived benefits and drawbacks of BWC use during the George Floyd protests (RQ3) and how these may vary by agency size (RQ4). The key benefits were evidentiary value and "civilizing effect," whereas drawbacks centered on insufficient battery life and mounting difficulties.

Benefits of BWCs during protests. The most common benefit was the evidentiary value of footage (52% overall; 42% small, 62% medium, 62% large). Many respondents pointed to the footage as a mechanism for providing clear evidence of what transpired:

	All (n = 75)	Small (n = 39)	Medium (n = 20)	Large (n = 16)	Chi-square
BWCs were present	87% (59/68)	86% (30/35)	82% (14/17)	94% (15/16)	1.002
BWCs were useful	78% (39/50)	71% (17/24)	85% (11/13)	85% (11/13)	2.526
Evidentiary value	52% (26/50)	42% (10/24)	62% (8/13)	62% (8/13)	
Behavior change	8% (4/50)	13% (3/24)	0% (0/13)	8% (1/13)	
BWC notification					3.868
Mandatory	21% (12/56)	33% (8/28)	7% (1/15)	20% (3/15)	
Recommended	32% (18/56)	33% (8/28)	47% (7/15)	20% (3/15)	
Not required	46% (26/56)	43% (12/28)	47% (7/15)	47% (7/15)	
BWC activation					1.601
Required – direct interaction	45% (25/56)	45% (13/29)	54% (7/13)	36% (5/14)	
Required – entire event	38% (21/56)	34% (10/29)	38% (5/13)	43% (6/14)	
Discretionary	16% (9/56)	21% (6/29)	8% (1/13)	14% (2/14)	
Limitations of BWCs					
Battery/recording capacity was sufficient	84% (46/55)	86% (25/29)	86% (12/14)	64% (9/14)	9.759*
No other technical problems	80% (45/56)	89% (25/28)	86% (12/14)	57% (8/14)	7.981
Complaints filed against officers at protests	22% (14/64)	0% (0/33)	24% (4/17)	67% (10/14)	29.383***

Note(s): $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$

Table 1. BWCs at protests

> Absolutely, a necessity for a police department. If agencies fail to implement BWCs they are reducing their own capacity to help deliver the narrative about an event. (Small agency)

> The BWCs were also useful in documenting officer's actions during enforcement efforts. (Large agency)

Respondents also highlighted the usefulness of BWCs for identifying suspects after-the-fact.

> Video captured by BWCs have been used to identify suspects during incidents of looting, rioting, assaults on public/officers. (Large agency)

> We were able to provide many hours of video to ATF arson investigators that showed rioters before and during the assault on our police station and at other riot locations, so that they could compare video images with other evidence in an effort to charge the arsonists. (Large agency)

These results suggest the evidentiary value of BWCs extends to protest policing.

We also searched for indications that agencies deployed BWCs to produce a "civilizing effect," operationalized as behavior change for either citizens or officers. Only a few agencies highlighted this potential benefit (8% overall; 13% small, 0% medium, 8% large). Several respondents also noted BWCs helped spark conversation with protestors or de-escalate situations.

> I like the capability to use for interactions when an unlawful act needs to be addressed. Additionally, it shows the public we are using them and it sparks conversations and sometimes de-escalation when agitators see them and realize they could potentially be recorded. (Small agency)

> Yes [BWCs are useful]. They changed the dynamic of the initial interaction with protesters in a very positive way. (Small agency)

However, most respondents did not recognize a civilizing effect as a benefit for using BWCs within the context of policing protests. To better understand this finding, we assessed whether the preconditions for a civilizing effect were met.

Of the four preconditions for producing a civilizing effect, one is captured above: The increased violence of the protests (e.g. higher mean scores) can be used as a proxy for the precondition of escalated behavior. We further assess two additional preconditions: Citizen awareness of the BWC and officer activation of the BWC. First, we examined the administrative policies of all responding agencies to determine whether they require citizen notification about BWCs (the most common mechanism for generating citizen awareness). Table 1 shows that overall, 21% of policies state that citizen notification is mandatory, 32% recommend it, and 46% do not require it. Mandatory notification was least common among medium-sized agencies (7%). Given that most departments either mandate or recommend notification (53%), this precondition for a civilizing effect is often, though not always, met (e.g. Patterson and White, 2021).

Second, we asked respondents about departmental policy on BWC activation during protests. Of the 56 agencies that responded to the question, the majority (83%) stated the activation is required, though agencies differ substantially on *when* activation occurs. About half (45%) reported that activation is required upon "interaction with the public," "when criminal behavior is present," or when officers are "dispatched to a call for service." These agencies do not record the entire protest, only specific interactions between officers and citizens. Alternatively, 21 agencies (38%) noted that activation is "required for the entirety of the event." Assuming officers activate as required by policy, the vast majority of agencies (83%) meet this pre-condition for a civilizing effect. The remaining agencies left the activation decision to the officers (i.e. discretionary; 16%) [10], as described by this respondent:

> The university experienced a protest that was peaceful. Officers working in a patrol capacity on the day of the protest did wear body cams as is the normal Department policy; however, the cams were never activated to capture any footage of the demonstrations. (Small agency)

Limitations of BWCs during protests. We were also interested in the perceived limitations of BWCs within the context of protests. Table 1 shows that the majority of respondents indicated that BWC battery life and recording capacity were sufficient (84%).

> Yes the batteries from [BWC vendor] are quite robust, and recording capacity is more than adequate even for extended situations such as civil unrest. (Large agency)

Large agencies were significantly less likely to report satisfaction with battery life and recording capacity (64% compared to 86% for medium-sized and small agencies; chi-square = 9.759; $p < 0.05$). This may be explained by the higher degree of reported incivility or longer protests in large agency jurisdictions—or both. The primary other issue reported by agencies involved problems with BWC mounting:

> The biggest issue we had was BWCs falling off of officers. This is an issue with the magnet security attachment that [BWC Vendor] uses. The only officers to use this set-up are officers that wear an internal vest. Most officers now use an external vest, which allows for the BWC to be mounted securely to the officer. (Medium-sized agency)

> Magnetic mounting options not sufficiently strong enough to prevent BWC from falling off uniform while engaged in vigorous physical activity or physical confrontation. (Large agency)

While both of these are common limitations of BWCs more generally, they are also important drawbacks when considering the use of BWCs in protests. First, compared to everyday police encounters, protests are typically long events. As described above, there is substantial variability in department policy for BWC activation and duration of recording during protests. As such, some jurisdictions may experience battery life concerns only within the context of longer events (e.g. protests), especially if they experience multiple events in quick succession. Second, it is a well-documented problem that BWCs dislodge or are knocked loose in close quarters. This is highly likely during contentious protests as officers and citizens engage physically. Combined, insufficient battery life and mounting difficulties can result in encounters not being recorded or poor-quality footage that is virtually unusable. These logistical problems can create inaccurate public perceptions that police are intentionally violating agency policy to conceal their own wrongdoing, which, in protests like those in 2020, would only further deteriorate already-fragile public trust.

Complaints against officers

Related to complaints during protests, as we assessed the frequency and nature of complaints filed against officers and whether BWCs were useful in their investigation. Table 1 shows 22% of responding agencies received complaints against officers, though this varied considerably by agency size: Two-thirds of large agencies compared to 0 and 24% of small and medium agencies, respectively (chi-square = 29.383; $p < 0.001$).

> We have some complaints of excessive force and likely some lawsuits pending from people claiming injury from officers' use of riot control devices. (Large agency)

> Excessive force such as rubber bullets, tear gas, pepper spray, etc . . . Also civil rights complaints regarding rights to protest. (Medium-sized agency)

> Arrest of media and curfew violators. (Large agency)

Of the 14 respondents that received complaints, eight (57%) highlighted the evidentiary value of BWCs as an investigative tool. Specifically, respondents noted the use of BWC footage to review complaints and use of force situations.

> Yes [BWCs were present and activated]. Much of the video is still under review, but there does not appear to be violations of the department policy. A few minor training issues have been identified as well. (Medium-sized agency)

> BWCs recorded incidents. Most notable incidents to make national news have resulted in officers being suspended, terminated and/or charged criminally. (Large agency)

Discussion

The death of George Floyd sparked worldwide protests against police use of force. Those protests were far more organized than past protests against police brutality, and officers at those protests were far more likely to have BWCs. This created an opportunity to assess whether the well-documented benefits of BWCs—namely, evidentiary value and a civilizing effect—extend to policing public protests. We used survey data from 100 law enforcement agencies to explore four research questions, and there are several notable findings.

First, three-quarters of responding agencies encountered protests related to the killing of George Floyd (RQ1). Though our sample is narrowly defined, the prevalence of protests mirrors more general reporting (Silverstein, 2021). Second, the vast majority reported that officers deployed to those protests were outfitted with BWCs (RQ2). This is perhaps not surprising given the study participants were federal BWC grant recipients, but the appropriateness of BWCs at First Amendment-protested demonstrations has been a source of debate (Ng, 2020). In fact, the ACLU opposes their use:

> We do not want police to use the cameras to collect video of peaceful protesters, which could chill the freedom people should feel to exercise their First Amendment rights. Indeed, we recommend — and many local policies include — provisions barring the police from recording events such as peaceful protest marches (Stanley, 2020).

The results here suggest that most law enforcement agencies have set aside those concerns in favor of recording police-citizen interactions at George Floyd protests. Moreover, just over three-quarters of respondents (78%) found BWCs to be useful at the protests, and relatively few experienced problems.

The widespread use of BWCs at these protests presents an opportunity to explore whether documented benefits (and limitations) extend to protests (RQ3). First, our results demonstrate strong support for the evidentiary value of BWCs at protests. Most respondents pointed to the power of BWCs to neutrally record what occurs (Gaub *et al.*, 2021a). This is potentially valuable both for criminal prosecutions of protestors and for investigating allegations of officer misconduct. Though complaints filed against officers at protests were rare, respondents again pointed to the value of BWCs for investigating them. The second perceived benefit of BWCs is the potential to improve both police and citizen behavior, measured here as reductions in use of force and citizen complaints. Fewer agencies reported deploying BWCs for this benefit, though the results suggest at least three pre-conditions for a "civilizing effect" were arguably frequently present (escalated behavior, citizen awareness, BWC activation; Patterson and White, 2021). Last, there were notable differences in findings by agency size (RQ4), particularly with regard to the prevalence and civility of protests and the perceived benefits and limitations of BWC use during protest events.

Our findings have several policy implications. First, BWCs can provide a valuable visual and auditory record of what is happening during protests, which often last several hours, involve substantial numbers of people, and have varying levels of coordination/organization. Footage, especially from multiple officers, can serve as important evidence when piecing together incidents for criminal investigations and internal agency accountability processes. That said, BWC footage is not immune from differing interpretations, which can cause or

exacerbate tensions between police and the public. Our findings suggest the benefits of BWCs during public protests may go beyond their evidentiary value and increase the overall sense of accountability and transparency among those present (Maguire, 2021).

Additionally, deploying BWCs to public protests can inhibit free expression and the right to protest, especially when peaceful (Stanley, 2020). In fact, "the use of extensive dataveillance mechanisms [such as BWCs] and coordination [among] law enforcement can have chilling effects on peaceful protests" (Gaub and Koen, 2021). However, our findings show that most agencies do not differentiate a policy for BWC use during large-scale demonstrations—nearly 50% of agencies indicated their officers were instructed to only record "law enforcement encounters" or "incidents with citizens," and another 16% left activation to officer discretion. Public protests are highly fluid, unique events, and standard BWC policy language on activation and use may not be sufficient to adequately address the complexity of those events. Agencies should carefully weigh the evidentiary and potentially civilizing benefits of BWC use during specific encounters against the negative, unintended consequences [11] of recording people engaged in constitutionally-protected behavior.

The current study does have limitations that warrant discussion. First, the survey sample is a non-random group of agencies using BWCs. Respondents are not representative of all American law enforcement agencies (especially by agency size), or even those funded through the BJA PIP program; thus, the results presented may not be representative of the experiences in other jurisdictions. Additionally, there is no comparison to agencies without BWCs, whose responses may have yielded useful information about how BWCs *could* have been beneficial (or not) in their situations. Third, the George Floyd protests occurred at the height of the COVID-19 pandemic when many jurisdictions had lockdowns, mask mandates, and/or curfews in place and prohibited large gatherings of people, even outdoors. It is unknown whether (or to what extent) this co-occurrence altered the nature and frequency of the protests, or police and citizen behavior during the protests. Fourth, we ask respondents to describe protest civility from their perspective, and their interpretations of civility may differ from those of the public. This difference may be even more pronounced given that the police (both as a monolithic group and individual departments) were the subject of the George Floyd protests. Finally, we were unable to compare protest-related use of force incidents and citizen complaints to those unrelated to protests. Future research should assess the complexities of these outcomes under different types of circumstances. These limitations aside, the findings provide an important snapshot of the role and usefulness of BWCs during the 2020 protests.

Conclusion

The George Floyd protests highlight the occasionally conflicting roles of the police. They are tasked with maintaining order among protesters at the same time they are required to observe and protect protesters' constitutional rights. Moreover, their very presence potentially agitates protesters when police actions are the subject of the protest. Our findings suggest some of the established benefits of BWCs can also be realized in public protests.

Notes

1. There were several cases of police abuse of authority against mostly minority citizens in 2020, but George Floyd's death in Minneapolis drew the most scrutiny (Silverstein, 2021). For the sake of simplicity, we refer to the Black Lives Matter/police reform-related demonstrations in summer 2020 as the George Floyd protests.

2. Police use of force is measured in a number of ways, including official reports of force documented by the officer (most common) and citizen complaints of force. To address concerns that official records undercount incidents of force (Logan and Ferguson, 2016), scholars have recently begun independently assessing force using BWC footage.

3. There are two other widely-studied areas of BWC research—officer perceptions and citizen perceptions of BWCs—that are outside the scope of the current study, and given space constraints, are not reviewed here. Other less-studied areas include BWCs' impact on procedural justice/legitimacy, investigations, and other criminal justice outcomes (e.g. court processes).
4. Procedural justice is the way in which police treat citizens and the fairness of the decisions that police make (Sunshine and Tyler, 2003). Prior research has identified four key components of procedural justice: citizen participation (being given the opportunity to state one's case), fairness and neutrality, dignity and respect, and trustworthy motives (Tyler and Huo, 2002).
5. Of 426 emails sent, 53 were returned because of a nonfunctioning email address (e.g. the original point of contact for the BWC grant no longer worked at the agency) yielding a final sample size of $N = 373$.
6. We did not provide specific definitions of key concepts such as "protest" or "peaceful" to avoid priming participants; rather, we wanted to understand their perceptions of the events occurring in their jurisdiction.
7. There is not unanimous agreement on the optimal classifications of agency size, though some (e.g. LEMAS) are more commonly used. The distribution of our sample required a more tailored scheme. For example, our respondents include only five agencies with fewer than 10 sworn officers and five with 1,000 or more. Our classification creates meaningful categories with a sufficient number of agencies in each.
8. It is possible that respondents may have responded to the same protests, given a significant number of agencies sent officers to protests in other jurisdictions. Moreover, some respondents may share jurisdictional boundaries (e.g. a local agency and sheriff's department in the same county). Given our research questions and space constraints, we did not explore this issue, nor did we compare responses between county and local agencies.
9. 59 of the 75 agencies that experienced protests responded to the question about officer injuries with "no" or "not applicable" (based on the previous question about civility). Two agencies that experienced protests did not respond to the injury question.
10. One agency reported that the policy was "unknown."
11. Examples of such consequences are inclusion in databases of "extremist" or terrorist individuals or groups, continued surveillance outside of defined protest events, and monitoring of social media accounts (Gaub and Koen, 2021).

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